

Effect of dielectric strength of mineral oil on mixing with a natural ester oil as a polar contaminant under AC stress

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Abstract- Ester oils are considered as suitable substitutes for conventional mineral oil for transformer applications due to good environmental performance and higher fire point compared to mineral oils. The electronegative property of the oxygen atom in the ester oil group makes ester oil more polar than mineral oil. Ester oils also have higher permittivity and lower resistivity with respect to mineral oil. Hence, it would be of interest to predict the influence of polar contamination in a non-polar nature of mineral oil. In this paper, 10% of natural ester oil is mixed as a polar contaminant in mineral oil to evaluate its effect on AC dielectric strength. Initially, AC dielectric strength of mineral oils and natural ester oil are predicted as a reference for various oil gaps under power frequency voltage conditions. Both naphthenic and paraffinic based mineral oil grade (uninhibited and inhibited oils) are utilized to consider the effect of inhibitors and aromatic contents of mineral oil in natural ester oil on AC dielectric strength. The breakdown voltages are determined by rising voltage method for nonuniform field (Sphere-needle) configurations. Weibull distribution functions are used to analyze the experimental data and to estimate the breakdown voltage with risk of 1% failure probability. Finally, finite element method (FEM) is used to predict the maximum stress in mineral oil, natural ester oil and mixture oil.

Keywords – Mineral oil, Natural ester oil, Statistical distribution, AC stress, Dielectric strength.

I. INTRODUCTION

Mineral oil is widely used in transformers as a dielectric medium and as a coolant due to its wide availability, good physical performances, electrical performances, reliability and low cost. Hence, significant experience has gained experimentally to determine the breakdown strength of mineral oil [1-7]. With increasing interest in natural ester oil as a substitute for mineral oil, an extensive work has been published to compare the physical and chemical properties of natural ester oils with mineral oils for transformers [6-9]. The chemical composition and molecular structure of natural ester oil is different from mineral oil [6-11]. The good solubility of water in natural ester oil makes it more insensitive to moisture than mineral oil. Due to these differences in physical and chemical properties, it is important to ensure that ester oils have adequate and acceptable dielectric strength in transformer applications [6,7].

Generally, natural ester oils are completely miscible and appear like transparent oil when mixed with mineral oils. Therefore, there is a possibility that the two different oils may get mixed with each other accidentally either on the

shop-floor while manufacturing the transformer or during service on site. With high proportion of natural ester oil in mineral oil, the performance of the mineral oil may degrade or improve asymptotically and it may change the electrical, chemical and thermal performance for transformer applications [6-14]. However, the fire point and biodegradability of the mixture oil (natural ester oil in mineral oil) will increase with respect to mineral oil [6, 7]. In this paper, dielectric strength is analyzed on a mixture of 10% of natural ester oil in mineral oil for various oil gaps between 1mm to 20mm in steps of 1 mm using sphere-needle electrodes under AC stress. The electric field distributions between sphere-needle electrodes are also calculated using FEM simulations based on 1% failure probability of AC withstand voltage using Weibull distributions for various oil gaps.

II. EXPERIMENTAL DISCRPTIONS

A. Transformer oil and test cell

Mineral oil is a mixture of various hydrocarbon molecules having various structures such as paraffinic, naphthenic and aromatic [6, 7]. Paraffinic based mineral oil consists of substantial quantities of naturally occurring paraffins than naphthenic oil. The oxidation of paraffin produces an insoluble sludge which leads to higher viscosity and pour point. Naphthenic oil has more polar features than paraffinic oil due to the presence of aromatic compounds which remain in the oil at low temperature [6,7,12,13]. Naphthenic oil is more readily oxidized than paraffinic oil. To improve the oxidation stability of oil, the mineral oils may also contain inhibitors.

Natural ester oil is derived from natural plants (vegetable seeds). The main molecular component of natural esters is Glycerol esters. The ester linkage has three different C18 fatty acid chains which will contain one, two or even three double bonds [6, 7, 15]. Hence, natural ester oil is more polar than mineral oils due to the presence of ester linkages. In this paper, Transol grade of mineral oil, produced by Savita Oil Technologies Limited, and BioTransol grade of natural ester oil produced by Savita polymers limited are used to compare and predict the AC dielectric strength of pure liquids (mineral oil and natural ester oil) and a mixture containing 10% natural ester in mineral oil.

The test cell (1L volume) is fabricated according to ASTM D-3300 [16]. The cell is made from transparent acrylic in order to observe the breakdown phenomenon. The experimental set-up is used to establish the experimental

confirmation and statistically supported value of the voltage of the oil at different intervals. Sphere-needle electrode configurations are chosen in this study to create non uniform local electric fields [3, 4, 12-15]. Steel phonograph needle is used as the point electrode having the tip radius of curvature 0.06mm ($\pm 20\%$) while the sphere electrode is made of brass having a diameter of 12.7mm. The gap distance between electrodes in the test cell can be set in steps of 0.5 mm with the help of screw gauges on either side of the electrode terminals.

B. Sample preparation

The first objective of this paper is to evaluate the breakdown voltage of mineral oil and natural ester oil and comparison of their breakdown voltages at a low failure rate (1% failure rate). Generally, the evaluation of 50% breakdown voltage is in fact of minor importance compared to the voltage range of low breakdown probability (1% failure probability), because only a sufficiently low breakdown risk is acceptable for technical dimensioning [17]. This objective demands high quality of oil preparation. Hence, the mineral oil and natural ester oil samples are directly taken from sealed barrels in order to maintain same consistency of oil quality for all the experiments. The properties of the oil (natural ester oil, mineral oil and its mixture oil) like breakdown voltage, water content and loss factor are evaluated prior to the test (Table I).

Table I. PROPERTIES OF PROCESSED SAMPLE OIL FOR EXPERIMENTAL ANALYSIS

Parameters of oil	Mineral oil	Natural ester oil	Mixture oil (90% Transol & 10% BioTransol)
	Transol	BioTransol	
Breakdown voltage in kV for 2.5mm gap	65-70	70-80	75-85
Water content in ppm	4-5	55-70	12-14
Relative permittivity	2.2	3.3	2.3
Viscosity at 40°C mm ² /sec	8.83	32	9.55
Relative saturation in %	10	7	7

A common procedure of sample preparation is used for both mineral oil and natural ester oil. The oil sample is filtered under vacuum using Whatman filter paper of size 0.45 μm . The filtered oil is heated at 100°C under vacuum oven at 720-740mmHg pressure to remove moisture content and dissolved gases from the filtered sample. As a general procedure, the heated oil is dehydrated and degassed in a vacuum oven before starting breakdown tests to minimize the influence of contamination in the test results [12, 13]. Therefore, after carefully filling with the oil sample, the test cell is kept still for about one hour before starting the breakdown tests, in order to remove the presence of any tiny air bubbles and stabilize oil samples.

The natural ester oil is mixed with mineral oil at ambient temperature with the aid of a magnetic stirrer at a speed of 1100 rpm for about 15min. To ensure the miscibility of mineral oil with natural ester oil, the mixtures were prepared and left for two days at rest. After that, a homogeneous mixture without phase separation is observed visually.

C. Testing method

AC voltage is supplied by single transformer rating 7.5KVA, 150kVrms, 50Hz. To measure the applied voltage a capacitive voltage divider is connected in parallel with the test cell. The needle electrode is connected to high voltage and sphere electrode to the ground. The gap distance between sphere- needle electrodes is varied from 1mm to higher values by a set of thickness gauges which is capable of increasing gap in steps of 0.5 mm. AC voltage is applied to test cell without any interruptions of applied voltage from zero to breakdown voltage with a rate of rise of 1-2kV ($\pm 0.2\text{kV/s}$). Each oil sample is tested with a series of 15 times to meet the amount of tests required by 1% failure probability analysis by using Weibull distribution.

The rising voltage method or step voltage method is generally used to measure the breakdown voltages [12-15] under AC stress. Hence, the following test procedure is followed for the measurement of breakdown voltage of each oil sample of mineral oil and natural ester oil and its mixture of oil.

- The initial voltage level is to be set at 60% of expected breakdown voltage
- Voltage is increased in steps of 2 kV and the test voltage is applied for the same length of time, usually 60 seconds
- The oil sample must withstand a minimum of three voltage levels before breakdown occurs.
- After breakdown occurs, there is atleast five minutes time interval before the next shot for natural esters and 11 minutes interval for mineral oil
- The oil is stirred after each breakdown and allowed to stand for a few minutes to avoid air bubbles.
- Oil sample is regularly changed after three breakdowns on average in order get consistent results.

III. EXPERIMENTAL RESULTS

The insulation strength of a transformer is predominantly decided by the strength of oil gap between the solid insulation [17]. Since, the solid insulation has higher dielectric strength than oil, the pressboard is used to subdivide the long oil gaps into smaller oil gaps [18, 19]. Therefore, the present study focuses on evaluating dielectric strength of oil for oil gaps of 1mm-20mm. The oils are compared in terms of breakdown voltage, standard deviation and maximum stress for the gap between the two electrodes of 0<1<20mm along the shortest field line in non uniform field.

A. Evaluation of Average value and standard deviation of breakdown voltage

The breakdown phenomenon of oil is a statistically distributed quantity which is dependent on the complex physics and chemistry of the oil and impurities involved [1, 2, 12-14]. The breakdown voltage of oil provides the useful information on the design criteria of transformer insulation systems and it would always obey probability distribution functions [13, 14]. Hence, the best estimates of the statistical parameters of average breakdown voltage and standard deviation of breakdown voltage are calculated initially for comparative study of natural ester oil, mineral oil and its mixture oil (Table II, III).

Table II. BREAKDOWN VOLTAGES OF THREE DIFFERENT GRADES OF MINERAL OIL AND NATURAL ESTER OIL

Statistical values	Gap in mm	AC voltage of mineral oil (kV)			Natural ester oil
		Naphthenic inhibited	Paraffinic grades		
			Uninhibited	Inhibited	
Average breakdown value in kV	1	21.4	22.7	21	15.4
	5	43.2	43.7	44	35
	10	63.5	63	63.3	43
	20	72	71.3	72	50
Standard deviations in kV	1	2.4	1.9	1.8	1
	5	2.6	1.4	1.4	1.7
	10	3	1.9	1.8	2.3
	20	3.1	1.8	1.7	2.3

Table III. BREAKDOWN VOLTAGE FOR MIXING OF THREE DIFFERENT GRADES OF MINERAL OIL WITH 10% OF BIOTRANSOL

Statistical values	Gap in mm	AC voltage under 10% natural ester oil mixture in to mineral oil (kV)		
		Naphthenic inhibited	Paraffinic grades	
			Uninhibited	Inhibited
Average breakdown value in kV	1	23	19	20
	5	49	38	41
	10	78	54	58
	20	88	64	66
Standard deviation in kV	1	1.7	1.6	1.5
	5	1.4	1.2	1.2
	10	1.7	1.6	1.4
	20	1.6	1.4	1.5

From Table II, the following points are observed

- The breakdown voltages of three different grades of mineral oil are same. The small difference in the breakdown voltage of different grades of mineral oil may be due to oil handling.
- In addition, the breakdown voltage of natural ester oil is less in comparison to all grades of mineral oil (naphthenic and paraffinic grades).
- Above 5mm oil gap, the standard deviation of natural ester oil is higher than paraffinic grades. However, naphthenic inhibited oil shows higher standard deviation than natural ester oil and paraffinic grades of mineral oil.

The natural ester oil and Naphthenic inhibited oil with the highest standard deviation implies that breakdown can occur at a very distant voltage value (greater or smaller) than its mean value. It is because in mineral oil, the breakdown is mainly induced only by positive streamers while the breakdown voltage of natural ester oil is induced by both positive and negative streamers [12-14]. However, the negative streamers can easily propagate in ester oil due to their high electronegativity.

From Table III, if the natural ester oil is mixed with different grades of mineral oil, the following points are observed.

- The breakdown voltage of naphthenic inhibited oil with 10% natural ester oil is increased with respect to pure naphthenic inhibited oil. It indicates that the presence of aromatics contents in the naphthenic oil is playing a major role with polar molecules of natural ester oil. The standard deviation of the naphthenic inhibited oil is lesser after 10% mixture of natural ester oil.
- The breakdown voltage of both paraffinic grades decreases when 10% of natural ester oil is mixed to it. However, the inhibited paraffinic oils show slightly higher breakdown voltage compared to the uninhibited paraffinic oil. It indicates that the inhibitors are also playing a role in the breakdown voltage besides polar molecules. The standard deviation of both paraffinic grades oil is marginally decreased after 10% mixture of natural ester oil.

B. Estimation of 1% failure probability

The prediction of withstand voltage is one of the important parameters for insulation design of transformer. It is generally recognized that Weibull distribution is the best suitable function to describe the breakdown voltage behaviour at low probabilities (i.e. high withstand probability) as well as to describe various types of failures rate than normal and Gumbel distribution [17]. Generally, normal distribution function tends to overestimate (larger prediction) the breakdown voltages than measurement results of breakdown voltage and Gumbel distribution tends to underestimate (smaller prediction) the breakdown voltage at low failure rate (<10%) and high failure rate (>90%) [12-14, 17]. In Weibull distribution, the measurement results of breakdown voltage overlapped well with the prediction values at the whole failure rate among all the expected distributions [17]. In addition, probability value (p-value) of Weibull distribution is the largest among all the expected distributions. Hence, Weibull distribution is effectively utilized in this paper to estimate the 1% failure probability of AC withstand voltage of mineral oil and natural ester oil for various oil gaps.

Generally, Weibull distribution estimates the statistical parameters based on the fitted curve with specific shape parameter (β), scale parameter (η) and threshold parameter (γ). The statistical parameter will be used generally to fit the breakdown data with different failure probability values. The application of three parameter Weibull distribution provides the good option to fit the breakdown voltage distributions of transformer oil.

In this paper, three parameter Weibull distributions is used to estimate the statistical parameters, which fit the breakdown data of natural ester oil and mineral oil with 1% failure probability values. The 3-parameter Weibull probability density function is given by

$$f(t) = \frac{\beta}{\eta} \left(\frac{t - \gamma}{\beta} \right)^{\beta-1} e^{-\left(\frac{t - \gamma}{\eta} \right)^{\beta}} \quad (1)$$

Where

$$f(t) \geq 0, t \geq \gamma,$$

$$\beta > 0, \eta > 0, -\infty < \gamma < +\infty$$

Three parameter Weibull distribution plots of the breakdown results are obtained with goodness of fit [12-14, 17]. Table IV shows 1% failure probability values of mineral oil, natural ester oil and mixtures of oils based on statistical parameters (goodness of fit) of specific shape, scale and threshold values.

From the Table IV, it is found that the withstand voltage of mineral oil is 67 kV for 20mm gap. However, the predicted result from Weibull distribution analysis provides an inevitable uncertainty [17]. Therefore, mineral oil withstand voltage might not be exactly 67 kV and it will be near the withstand voltage value of 60 kV. Hence, the actual withstand voltage lies between 60kV to 75kV based on 95% confidence interval. Therefore, it is understood from Table IV that the withstand voltage of mineral oil is 67kV and 95% confidence interval of withstand voltage level lies between 60kV to 67kV. Similarly, the values for other different grades of the oil are given in Table IV.

From the predicted results of Weibull distribution, 99% withstand voltage (1% failure probability) of mineral oil, natural ester oil and mixture oil are extracted from Weibull distribution analysis and it is shown in “Fig 1”. If the transformer designer needs to consider the confidence interval for insulation design criteria then it is reliable to utilize the lower level in the confidence limits instead of the predicted withstand voltage [14].

TABLE IV. 95% CONFIDENCE INTERVAL OF WITHSTAND VOLTAGE

Transformer oil		Gap in mm	95% confidence interval in kV		
			Lower boundary	1% failure probability	Upper boundary
Mineral oil (all three grades)		1	12	18	20
		5	36	41	44
		10	46	54	57
		20	60	67	75
Natural ester oil		1	8	14	18
		5	24	28	32
		10	36	38	43
		20	39	48	49
Naphthenic oil & 10% Natural ester oil		1	12	17	18
		5	40	45	47
		10	70	76	78
		20	79	83	88
Paraffinic grades & Natural ester oil	Uninhibited & 10% Natural ester oil	1	14	17	20
		5	32	34	37
		10	45	49	52
		20	58	60	63
	Inhibited & 10% Natural ester oil	1	14	17	21
		5	34	35	37
		10	52	55	58
		20	55	61	63

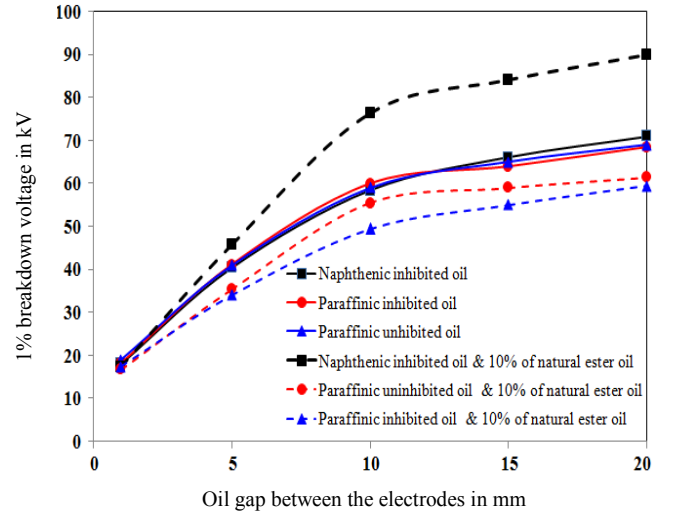


Fig 1. 1% failure probability of breakdown voltage for different grades of mineral oil with 10% of natural ester oil

IV. ELECTRIC FIELD DISTRIBUTIONS

Under AC stress, relative permittivity plays a major part in electrical field distribution. The maximum electrical stress is one of the major factors which determine the insulation failure of any insulation systems [18, 19]. For optimal design of any insulation systems and to achieve good safety margin it is important to understand the electric field distributions in the oil gap between the electrodes due to breakdown voltage [19].

Generally, performance of the oil mainly depends upon the field configurations, relative permittivity and its magnitude. The relative permittivity of ester oils (3.3) is higher than mineral oil (2.2). In mineral oil, maximum stress is experienced by oil and solid insulation barriers (pressboard and paper) will get lesser stress. However, in natural ester oil, the solid insulation barrier gets higher stress compared to solid insulation in mineral oil. Since, the permittivity of natural ester oil is close to solid insulation of pressboard (4.4), and it results in more even distribution of stress in natural ester oil compared to different grades of mineral oil [1,2]. Therefore, the electric field distribution (maximum stress) due to breakdown voltage of natural ester oil, mineral oil and the mixture of oil is also calculated (“Fig 2”).

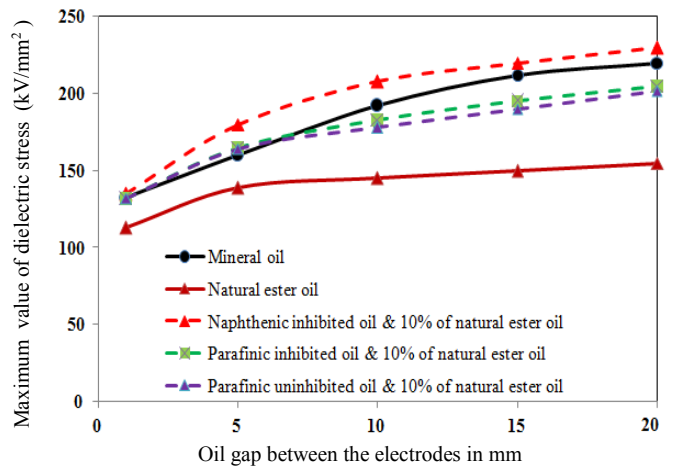


Fig 2. Maximum electric stress in the different transformer oil versus gap between the electrodes

FEM technique is used and it utilizes three dimensional Newton Raphson methods to calculate the electric field distribution between the two electrodes [17, 19]. The evaluation method compares the average stress for the whole range of 1 to 20mm gap between the electrodes along the shortest field line in the non uniform field. The variation of maximum electric stress (E_{max}) is shown for three oils due to change in permittivity and 1% failure probability voltage.

V. APPLICATION OF ESTER OIL FOR TRANSFORMER IN DESIGN IMPLICATIONS

During insulation design analysis, 99% withstand voltage of transformer oil should always be used with a sufficient safety margin between maximum stress and withstand strength to fulfill the safety requirements at the time of manufacturing. Hence, a reliable comparison between AC breakdown strength of different grades of mineral oil with 10% mixing of natural ester oil is predicted. From experimental analysis, the ratios of breakdown voltage & maximum electrical stress of mixture of oils and mineral oil under power frequency test conditions are calculated and its results shown in “Fig 3” and “Fig 4”.

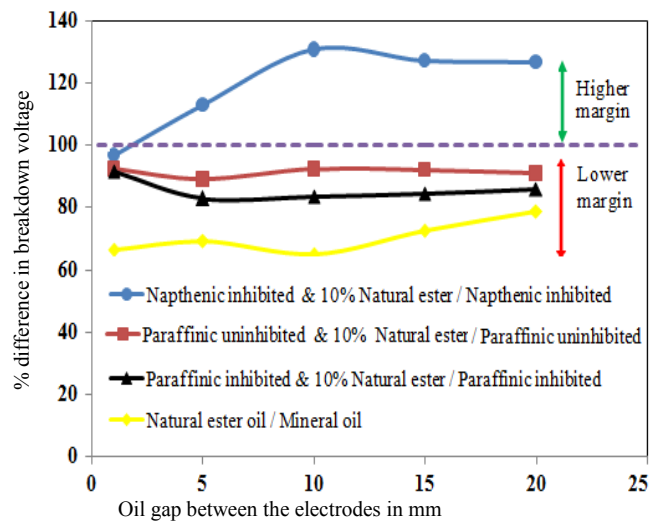


Fig 3. Effect of mixing of natural ester oil in mineral oil in terms of breakdown voltage.

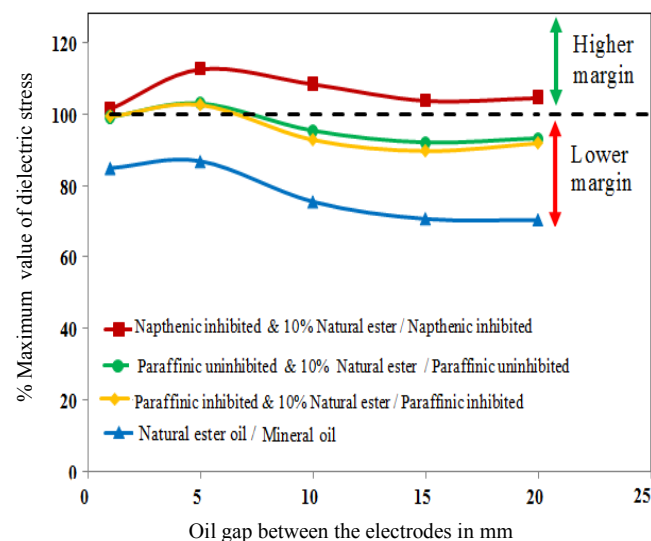


Fig 4. Effect of mixing of natural ester oil in mineral oil in terms of maximum stress.

From “Fig 3”, withstand strength of natural ester oil is about 65-78% of mineral oil. Natural ester oil has lower safety margin with respect to mineral oil. Hence, natural ester oil transformer design insulation clearance needs to be enhanced with respect to mineral oil transformer under AC stress. If natural ester oil is mixed with paraffinic grades of mineral oil, then the breakdown voltage is reduced and it is about 90-95% of pure paraffinic grades of mineral oil. To compare the electrical gradient of natural ester oil with mineral oil due to 99% withstand voltage for electrical design limits predictions, electrical field analysis is performed in finite element method based software. The effect of breakdown voltages of different oils is clearly reflected in maximum stress value and it shown in “Fig 4”.

Due to reduction of breakdown voltage of paraffinic grades of mineral oil after 10% of mixture of natural ester oil, the strength of dielectric stress is decreased and it leads to reduction of safety margin with respect to pure mineral oil. Although the breakdown voltage of both paraffinic grades of mineral oil mixed with natural ester oil are slightly lower (about 90-95%) than those of base pure mineral oil, it cannot be regarded as similar breakdown voltage when considering the safety margins used in transformer insulation design. It indicates that, if the natural ester oil is mixed with paraffinic grades of mineral oil accidentally or intentionally during manufacturing or at service depending on the percentage of the natural ester oil in the mineral oil transformer may fail at test bed or at site conditions depending on the safety margin provided in the pure mineral oil transformer guidelines.

VI. DISCUSSIONS AND CONCLUSIONS

With growing interests to use natural ester oil as alternative to petroleum based mineral oil for transformer application, an experimental analysis is performed to predict the withstand voltage of natural ester oil in comparison with conventional mineral oil. In addition, behavior of different grades of mineral oil when mixed with 10% of natural ester is also analyzed for transformer applications.

The breakdown voltages of three different grades of mineral oil are same for the entire oil gap between the sphere-needle electrodes. The breakdown voltage of natural ester oil is less with respect to all grades of mineral oil due to electronegative property in the natural ester oil. This is because the negative streamers can easily propagate between the electrodes due to high electronegativity in natural ester oil. During the experimental studies it is found that the condition of mineral oil after breakdown is pretty bad in comparison to natural ester oil in all oil gaps because of formation of carbon particles.

Additionally, when natural ester oil is mixed with naphthenic inhibited mineral oil, breakdown voltage increases in comparison to pure naphthenic inhibited oil which is an advantage for power industries from insulation point of view under AC stress. Hence, the presence of aromatic contents with inhibitors (polar molecules) in the naphthenic inhibited oil plays a major role to increase the breakdown voltage under AC stress. In addition, the dielectric breakdown voltage of paraffinic grades decreases after the addition 10% of mixture of natural ester oil in comparison to pure paraffinic grades. Thus, a significant conclusion of this study is that from the dielectric strength

point of view under AC stress, the mixtures of natural ester oil constitute an option for improving the dielectric performance of naphthenic grade inhibited mineral oil and but not in the case of paraffinic grades of mineral oil.

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